DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Supplementary Examination 2022

Course: B. Tech. Branch : ELECTRONICS AND TELECOMMUNICATION ENGINEERING

Semester : IV (PATTERN 2017)

Subject Code & Name: BTBSC406 Numerical Methods & Computer Programming

Max Marks: 60

Date: **Duration: 3.00 Hr.**

Instructions to the Students:

1. *Solve any **FIVE** questions*
2. *Draw neat diagram wherever necessary.*
3. *Figures to right indicates full marks*
4. *Assume suitable data wherever necessary and mention it clearly*

		Marks
Q. 1	**Solve the following.**	
A)	Find absolute error if the number X=0.00599826 is i) Truncated to four decimal digits ii) Rounded off to four decimal digits	**06**
B)	If $X^2 - e^{-X} = 0$, find real root by Newton Raphson method	**06**
Q.2	**Solve the following.**	
A)	Solve the following by Gauss elimination method 2x+y+z = 10 3x+2y+3z =18 X+4y+9z = 16	**06**
B)	Define Errors. Explain different types of errors occur in numerical computations	**06**
Q. 3	**Solve the following.**	
A)	Given the values X : 5 7 11 13 17 F(X) : 150 392 1452 2366 5202 Evaluate f(9) using i) Lagrange's formula ii) Newton's divided difference formula	**12**
Q.4	**Solve the following.**	
A)	Apply Runge kutta 4th order method to find approximate value of y for x =0.2 in steps of 0.1, if $dy/dx = x+y^2$ given that y=1 where x =0.	**06**
B)	Employ Taylor's method to obtain approximate value of y at x =0.2 for the differential equation dy/dx =2y + 3e^x, y(0) =0	**06**
Q. 5	**Solve the following.**	
A)	Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using i) Trapezoidal rule ii) Simpson's 1/3rd rule iii) Simpson's 3/8th rule	**12**
Q.6	**Solve the following.**	
A)	. Apply Lagrange's formula inversely to obtain a root of the equation f(x)=0, given that f(30)= -30,f(34)= -13, f(38) =3, f(42) =18	**06**
B)	Solve the following equations by LU decomposition (Factorization method) 2x+3y+z = 9 X+2y+3z = 6 3x+y+2z =8	